O_n as Auditable Ledger

Telemetry Integrity for State-Contingent Supply Control

Companion paper to the PDM whitepaper | Mann Mechanics Ltd | April 2026

PDM is a non-discretionary supply control system. This framework defines the conditions under which its inputs can be considered real.

Abstract

Progressive Depletion Minting (PDM) provides a deterministic, state-contingent control law for bounded resource pools, governed by measured obligations (O_n) and system velocity (V_n). The formal proofs of the PDM control law establish that, given authentic inputs, supply cannot be expanded arbitrarily. However, the control law does not itself solve the oracle problem. If O_n can be manipulated at the point of measurement, the mechanism degrades to garbage-in, garbage-out.

This paper specifies a telemetry integrity framework for O_n that closes this gap. The framework defines a fixed ontology of obligation types, a dual-key validation protocol, temporal expiry rules, netting and deduplication procedures, cryptographic attestation, separation of powers between fiscal authorities and obligation validators, and a constrained override protocol for genuine emergencies. The combined effect is that manipulation cannot be absorbed into the issuance rule. Any deviation must occur as an explicit, bounded, time-limited, publicly logged exception.

The framework is presented as a specification. It does not constitute a legal or regulatory instrument. Sector-specific performance claims are flagged as unproven in the absence of real-world deployment data.

1. The Oracle Problem as the Remaining Attack Surface

The PDM control law, as specified in the PDM whitepaper, is unconditional given valid parameters and valid telemetry. Theorems 1 through 5 establish non-negativity, boundedness, conditional minting, progressive resistance, and auditability. None of these proofs make claims about the authenticity of the telemetry inputs themselves.

This is deliberate. A control law and an oracle are distinct engineering concerns. Conflating them produces systems that are either mathematically elegant but operationally fragile, or operationally robust but mathematically opaque. Separating them makes both tractable.

However, the separation creates a clear attack surface. If O_n is computed from inputs that can be manipulated, then even a perfectly proven control law will expand supply in response to false signals. Three failure modes deserve explicit treatment.

1.1 Discretionary redefinition

A sovereign or institutional actor declares that a new category of claim counts as an obligation for the purposes of O_n. Future fiscal commitments, political promises, or speculative liabilities are added. The control law then mints supply against obligations that have no verifiable counterparty.

1.2 Unilateral insertion

A single actor with write access to the obligation register inserts synthetic entries. Without independent corroboration, the control law treats these as genuine and expands accordingly.

1.3 Temporal manipulation

Legitimate obligations are timed to coincide with advantageous positions in the control cycle. Expiry is deferred indefinitely. The result is a permanent inflation of O_n that never clears, producing sustained minting without corresponding contraction.

Each of these modes is independent of the control law. Each requires a telemetry-layer response.

> **Telemetry integrity is the system boundary. Outside it, guarantees do not hold.**

2. Fixed Ontology of Obligation Types

The first requirement is that the categories of claim admissible as components of O_n are finite, predefined, and enumerable. Redefinition is possible, but only through an explicit governance process that is itself logged and auditable.

For a monetary deployment, the admissible categories are restricted to legally enforceable, third-party verifiable claims. Illustrative examples include:

- Unpaid invoices with counterparty confirmation and a due date within the current accounting period.
- Payroll obligations with corresponding employment contracts and statutory filings.
- Tax payments assessed and due under the relevant jurisdiction's statutory calendar.
- Settlement backlogs in recognised clearing systems, with matching trade confirmations on both sides.
- Contractual delivery obligations with signed counterparty agreements and recorded performance triggers.

The ontology explicitly excludes discretionary fiscal projections, speculative liabilities, unfunded political commitments, and any claim without a named, solvent, legally identified counterparty. These exclusions are the operative constraint. They are what prevents the mechanism from being weaponised by a state that wishes to inflate O_n in order to justify issuance.

> **Illustrative adversarial scenario.** A government proposes to classify projected future spending commitments as obligations in order to expand O_n and justify issuance. The proposed entries name no solvent counterparty, have no statutory due date, and carry no independent validator signature. They fail admission at the ontology layer before reaching the validation layer. The attempt is rejected. Any override of this rejection must be an explicit, publicly declared, time-limited intervention under the constrained override protocol in Section 8.

Sector-specific deployments will require sector-specific ontologies. A carbon credit deployment will enumerate verified emission obligations, retirement commitments, and compliance deadlines. An energy grid deployment will enumerate contracted load, reserve margin commitments, and interconnection obligations. The general principle is unchanged: the set is finite, enumerable, and governed by explicit revision procedures.

3. Dual-Key Validation

Every entry in the obligation register must be confirmed by at least two independent sources before it is admitted to O_n. The sources must be organisationally and technically independent. Two systems operated by the same authority do not constitute independence.

For a monetary deployment, examples of qualifying source pairs include:

- A commercial bank's counterparty confirmation paired with a clearing system's settlement record.
- A tax authority's assessment paired with the obliged party's filed return.
- A payroll provider's disbursement schedule paired with the recipient's identity confirmation through a separate channel.

The validation requirement is not merely procedural. It is cryptographic. Each source signs a structured attestation that includes the obligation identifier, the amount, the counterparty, the due date, and a timestamp. The two signatures are combined into a validation proof that is stored alongside the obligation entry. If either signature is absent or invalid, the entry does not enter O_n.

This structure forecloses unilateral insertion. No single actor, however senior, can add to O_n without a corroborating attestation from an independent source.

4. Temporal Expiry

Every obligation entry carries an explicit expiry. If the obligation is not settled, renewed, or formally restructured before expiry, it is removed from O_n automatically. This prevents the accumulation of stale obligations that inflate the denominator of the supply-to-obligation ratio indefinitely.

Expiry periods are set by ontology category. Short-cycle obligations such as daily settlement claims expire within days. Medium-cycle obligations such as invoices expire within the standard commercial payment terms plus a bounded grace period. Long-cycle obligations such as tax liabilities expire at the end of the statutory collection window.

Renewal is permitted but requires a fresh dual-key validation. A renewed obligation is, for audit purposes, a new entry with a clear link to its predecessor. This prevents a single original entry from being extended indefinitely through silent rollover.

5. Netting and Deduplication

Obligations that are mutually offsetting, circular, or duplicative must be netted before they enter O_n. Three patterns require explicit treatment.

5.1 Bilateral netting

When party A owes party B a sum, and party B simultaneously owes party A a smaller sum, only the net difference is admitted. This prevents the artificial inflation of O_n through reciprocal contracting between related entities.

5.2 Circular netting

When a chain of obligations returns to its origin (A owes B, B owes C, C owes A), the circulating portion is netted out. Only the residual difference at each node remains.

5.3 Deduplication

When the same economic obligation is recorded in multiple systems (for example, an invoice recorded both in the vendor's receivables and the customer's payables), the duplicate entries are collapsed to a single obligation. Cryptographic hashing of the canonical obligation description provides the deduplication key.

Netting and deduplication are run as a scheduled process on the obligation register before O_n is computed for the next control cycle. The netting results are themselves hash-chained, producing an audit trail of how the aggregate O_n was derived from the raw register.

6. Cryptographic Attestation and Hash-Linked Reporting

Every state change in the obligation register is cryptographically attested and linked to its predecessor through a SHA-256 hash chain. The attestation covers the entry content, the validating signatures, the timestamp, and the previous chain head. The chain itself is published in a form that permits independent verification by any party with access to the public head.

This produces three operational properties. First, tampering with historical entries is detectable, because any modification invalidates the chain from the point of tampering forward. Second, the provenance of the aggregate O_n used in each control cycle is reconstructible from the chain. Third, external auditors can verify the integrity of the telemetry without requiring privileged access to the underlying systems.

The hash chain is distinct from the PDM control law's own audit chain. They are complementary. The control law chain records the mechanism's internal state transitions. The telemetry chain records the evolution of the inputs from which those transitions are computed. Both are required for end-to-end auditability.

7. Separation of Powers

The authority that benefits from issuance must be structurally separated from the authority that validates the obligations driving issuance. In a monetary deployment, this means the fiscal authority cannot be both the source of O_n data and the beneficiary of minted supply.

The specification requires three distinct roles.

- Obligation originators: commercial and institutional parties who record legitimate claims in their own systems.
- Obligation validators: independent entities that verify originator submissions and produce the second signature in the dual-key protocol. Validators must be structurally separate from originators, from the fiscal authority, and from any recipient of minted supply.
- Fiscal authority: the party that operates the monetary system and is subject to PDM's constraints on issuance.

Separation is not merely an organisational preference. It is a precondition for the framework's integrity. If any one actor controls two of these roles, the dual-key protocol collapses and O_n becomes manipulable through internal collusion. Jurisdictions that cannot enforce this separation in law should not deploy PDM as a monetary system. The framework will not protect against a unified actor that refuses to be bound.

8. Constrained Override Protocol

No control system survives contact with genuine emergencies if it has no override capability. The question is not whether overrides exist, but whether they are transparent. The specification defines a constrained override protocol with four properties.

8.1 Explicit

An override is invoked by a named authority through a publicly recorded declaration. It cannot occur silently. The declaration states the specific parameter being overridden, the magnitude of the deviation, the justification, and the expected duration.

8.2 Bounded

Override authority is limited by prior rule to a maximum deviation from the mechanism's own output. A deviation exceeding the bound requires a higher tier of authorisation with correspondingly greater public visibility. Unbounded overrides are not permitted.

8.3 Time-limited

Every override has a predefined expiry. At expiry, the mechanism reverts automatically to its rule-governed output. Renewal requires a fresh declaration, not a silent extension.

8.4 Logged

Every override is entered in the control law's audit chain with the same cryptographic discipline as any other state transition. A user of the system, inspecting the chain, can distinguish rule-governed transitions from override transitions at every point in history.

The effect of this protocol is that emergency powers remain available, but they become legible. Political and economic capture that relies on hidden intervention is foreclosed. Capture that occurs through openly declared, repeated override remains possible but is now visible to counterparties, citizens, and external auditors.

9. Boundary Conditions

The framework is designed for jurisdictions operating under rule of law. Three specific preconditions must be satisfied for the specification to deliver its intended guarantees.

- Dual-key validation must be legally enforceable. Courts and regulators must recognise the cryptographic attestations as admissible evidence of obligation existence and magnitude.
- The fixed ontology must be embedded in statute or equivalent instrument. A government retaining the ability to unilaterally redefine admissible obligation types through executive action defeats the ontology's purpose.
- The separation between fiscal authority and validators must be structurally real. Nominal separation without operational independence is not sufficient.

In jurisdictions where these preconditions cannot be satisfied, the framework will not provide its stated guarantees. This is a boundary condition, not a flaw. No technical or mathematical system can substitute for a state's willingness to be bound by its own rules.

10. Reference Architecture: Monetary Deployment

A monetary deployment of PDM with the full telemetry framework comprises the following components.

- An obligation register, operated by a chartered institution independent of the fiscal authority, recording all admitted entries with their dual-key validation proofs and hash chain.
- A validator panel composed of statutorily independent entities: for example, a national audit office, a central clearing counterparty, a tax administration operating under judicial review, and one or more commercial auditing bodies rotated on a fixed cycle.
- A control law instance operating PDM's burn, conditional mint, and progressive damping functions against the aggregate O_n produced by the register at each cycle.
- A publication layer that makes the control law audit chain and the telemetry audit chain available for independent verification.
- A constrained override authority exercisable only by named officers under statutory conditions, with mandatory public declaration and time-limited effect.

The specification does not prescribe the distribution of minted supply. Distribution is a separate policy layer that must be designed to address the Cantillon problem explicitly. PDM with telemetry integrity removes discretionary timing and quantity of issuance; it does not by itself determine who receives new supply first.

11. Reference Architecture: Carbon Credit Deployment

Carbon markets present a structurally analogous problem to monetary systems. The supply of credits must be bounded by a verifiable cap, expansion must respond to genuine compliance obligations, and manipulation of the underlying obligation measurement has historically undermined confidence in the asset class.

A carbon credit deployment of PDM with the telemetry framework comprises the following components.

- An obligation register of verified compliance liabilities: regulated emitters' reported emissions, retirement commitments, and compliance deadlines.
- A validator panel of independent verification bodies operating under the relevant accreditation regime, cross-checked against regulator filings.
- A fixed ontology restricting admissible obligations to regulated compliance instruments with identified liable parties and statutory due dates.
- A control law instance governing the issuance of new credits against the cap, with progressive resistance as the cap is approached.

- Hash-chained publication of both the obligation register state and the control law transitions, enabling independent market participants to verify that issuance is responsive only to authentic compliance demand.

Sector-specific performance claims for this architecture are unproven in the absence of real-world deployment. The specification is presented as a candidate design, not as a validated production system.

12. Closing

The PDM control law makes manipulation of supply impossible given authentic inputs. This companion specification makes the inputs themselves auditable. The two together constitute a complete supply control system for jurisdictions willing to operate under rule of law.

The framework's contribution is not the individual components. Dual-key validation, hash-chained audit, separation of powers, and constrained overrides all exist in adjacent fields. The contribution is the assembly of these components into a coherent specification that closes the oracle problem for state-contingent supply control, and that does so without relying on the good faith of the parties being constrained.

Every previous monetary system permitted manipulation to be absorbed into the issuance rule. This framework, combined with the PDM control law, forces manipulation into the open. That is the structural improvement.

Disclaimer

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